



SURYA AI · TRADEPILOT · ENGINEERING AUDIT

Engineering Session Report

Root-cause diagnosis · A/B forward-test harness · ops hardening · dashboard rebuild

Period 2026-06-05 → 2026-06-07
System TradePilot autonomous trading fleet (v4 · v5 · v5_classic)
Author Surya AI · soumya@suryaai.co.in
Classification Internal · audit record
Generated 2026-06-07

§0 Executive Summary

This session diagnosed why TradePilot stopped making money, built a faithful forward-test to validate the fix, hardened the market-open automation, and rebuilt the live dashboard. The headline: the collapse was an **engineering regression, not a market**, and the simple pre-regression model is winning the live A/B.

AREA	OUTCOME	STATUS
v4 root cause	May-4 overfit retrain (5-tree → 1,735-tree) broke a working engine; old model wins A/B by +₹25,220 Day-1	DIAGNOSED
v5 root cause	Fair-weather long engine; short arm has no edge in bear (the regime it deploys most)	DIAGNOSED
A/B harness	Isolated forward-test of both challengers, automated via launchd	LIVE
Ops reliability	launchd exit-78 fixed & verified; wake + caffeinate + hotspot wired; Rust dropped	HARDENED
Dashboard	FLEET COMMAND sci-fi /live on real data; unified nav; A/B page; market-state sync	SHIPPED
Monday readiness	Fully automated if the Mac is awake at 09:10 (laptop-wake the only residual risk)	1 DEP

BOTTOM LINE

Neither engine has a robust all-weather edge — but v4's loss was self-inflicted (an overfit retrain) and is reversible. The decision to revert v4 to the 5-tree model awaits 2–3 clean A/B sessions; the CEO-override freeze expires **2026-07-15**.

§1 Root-Cause Diagnosis

1.1 v4 – the May-4 overfit retrain

v4 was gross-profitable through early May, then went flat-to-negative. A controlled comparison holding the *market regime constant* ruled out the market:

REGIME	OLD ENGINE (PRE-MAY-4)	NEW ENGINE (POST-MAY-6)
SIDEWAYS	+ ₹6,520 / day	– ₹5,846 / day
BEAR	+ ₹12,993 / day	+ ₹1,392 / day (9× worse)

The culprit (commit `1d174bc`, May 4): the ML model was retrained from a **5-tree** LightGBM into a **1,735-tree** model on the *same 22 features* — a 350× complexity explosion. Real-trade quality collapsed in step: win rate 61%→38%, per-trade edge + ₹141→– ₹8, hold time 97→172 min. Classic overfitting.

BIAS CAUGHT

An early "clean controlled experiment" leaned on 3–4-day samples (the project's own *don't-rush-findings* rule). The robust evidence is structural — the 350× tree explosion — not the small P&L windows. May-6's + ₹196k was largely a NaN/cache price artifact, not a real day.

1.2 v5 – a fair-weather long engine

v5's code never changed (Apr-16 → May-3 identical), so its decline was market + positioning, not a regression. Its edge is **long-heavy harvesting of market up-days**; its short arm is regime-conditional:

V5 SHORT ARM	TRADES	AVG / TRADE	WIN RATE
SIDEWAYS	217	+ ₹166	59%
BEAR	416	+ ₹3	39%

The engine deploys shorts *most* in bear (416 vs 217) — exactly where they don't work. Even in bear, v5's longs (+ ₹103/trade) beat its shorts (+ ₹3). Fix path: gate shorts out of bear, or run long-only.

1.3 The disk-full incident

Mid-session the machine was at 100% / 686 MB free — the live engine's writes were failing (empty cache dirs). Freed 2.2 GB by removing regenerable Rust build output (`engine/target`), likely a contributing factor to stale-cache symptoms.

§2 A/B Forward-Test Harness

A faithful historical backtest was **not possible** — the engine's intraday decision inputs were never persisted and there is no v4 backtest harness. The only honest validation is a live forward A/B, so two isolated challenger engines run alongside the live fleet:

- **v4 challenger** — the pre-May-4 5-tree engine (`tradepilot-oldengine-ab/`), with a ported NaN-price guard + a 09:30 warm-up guard for apples-to-apples timing.
- **v5 challenger** — current v5 with shorts disabled (`V5_LONG_ONLY=1` , `tradepilot-v5-longonly-ab/`).

Day-1 result (2026-06-05, clean session)

EXPERIMENT	CHALLENGER	BASELINE (LIVE)	DELTA
v4 model	+ ₹21,171 · 72% WR	− ₹4,049 · 41% WR	+ ₹25,220
v5 short arm	− ₹8 · 54% WR · gate ok	− ₹1,902 · 42% WR	+ ₹1,894

READING

v4 is a strong, clean signal (capital-normalized +0.78% vs −0.18% — not leverage). v5 is near-breakeven / inconclusive (one confounded day: regime mismatch + 15 unrealized open positions). Decision needs 2–3 more clean sessions.

§3 Operational Reliability

The worst recent P&L days were not trading failures — the engine simply **didn't launch on time** (the Mac failed to wake / launchd failed to fire). Fixes:

ISSUE	ROOT CAUSE	FIX
launchd exit-78	flaky <code>bash -c</code> under launchd; logs under <code>~/Documents</code> (TCC)	direct <code>python3</code> invocation + logs to <code>~/Library/Logs</code> — engines-on, preflight, exec-eod all now exit 0; 09:10 fire verified via kickstart
Mac not waking	<code>pmset</code> repeating wake self-clears	re-set <code>wakepoweron MTWRF 08:45</code> ; residual laptop limitation → plug-in + lid-open
processes sleep-killed	idle sleep killed <code>nohup</code> engines mid-morning	<code>caffeinate</code> owned by <code>launch-market.sh</code> through the session
network drop	no auto-reconnect	wifi-watchdog wired in → falls back to hotspot "Pro"
A/B not automated	only live engines in launchd	new <code>com.tradepilot.v2.ab-engines</code> job (09:12 weekdays → <code>launch-ab.sh</code>)
Rust execution layer	extra failure mode, low value	dropped — made optional; Python runs solo

RESIDUAL RISK

The one link that can't be made bulletproof on a laptop is wake-from-sleep. Durable fix = move the fleet to an always-on box (cloud VM / Mac mini) before the 2026-07-15 override expiry.

§4 Dashboard Rebuild

The live dashboard (`:5050/live`) was rebuilt into a sci-fi tactical command interface — **FLEET COMMAND** — wired to real data, with design choices grounded in verified research (Emil Kowalski's motion principles; "taste = restraint"; impeccable execution).

- **/live** — engine array with equity-curve sparklines · tactical radar (engine pods + orbiting trade blips + target-locks + fleet-P&L reactor core) · per-stock telemetry detail · 1-week session browser.
- **Craft from research** — `tabular-nums` so live numbers don't jitter (the #1 trading-UI fix); Emil's exact easing curves; staggered one-shot entrance; restraint — *no* animation on the 5s refresh.
- **Unified nav** — a top-right switcher (Desk · Market · Live · A/B · Agents) on every dashboard; market-state logic (OPEN only weekday 09:15–15:30 IST) synced across `/live` and `/lab`.
- **A/B page** (`/lab`) — the challenger-vs-live experiments in-app, with the long-only gate badge.
- **Security** — a reflected-XSS via the `?date=` param (date-note + fetch URL) was found by automated review and fixed by strict `YYYY-MM-DD` validation at intake.

A standalone multi-tab **vision mockup** (Desk/Market/Live/Agents reimaged) was also produced for design exploration, kept separate from production.

§5 Monday Readiness & Next Steps

Monday 2026-06-08 – automated sequence

08:45 IST	pmset wake (the one unreliable link → plug in + lid open)
09:10	engines-on → launch-market.sh: caffeinate + wifi→"Pro" + 3 live engines + dashboard
09:12	ab-engines → both A/B challengers (caffeinated)
~09:20	verify: /live MARKET=OPEN + trades populate; /lab cards fill
15:31 / 15:35	EOD reports + shutdown; read A/B Day-2 deltas

VERDICT

READY — the full stack fires hands-off provided the Mac is awake at 09:10. All launchd jobs exit 0 and the fire is verified.

Decisions & open items

ITEM	OWNER / TRIGGER
Revert v4 to the 5-tree model	after 2–3 clean A/B sessions confirm the +₹25k edge holds
v5 short arm — gate out of bear or run long-only	after a cleaner v5 A/B (fix regime mismatch + force-close)
Move fleet off the laptop (always-on box)	before 2026-07-15 override expiry — ends the wake-risk class
Restore Rust risk layer (optional)	only if Python-only risk gates prove insufficient

Artifacts & backups

docs/MONDAY_SPEC.md · docs/DESIGN_CRAFT.md · docs/V4_ENGINE_PORT_SPEC.md · scripts/launch-ab.sh · prototype/templates/{live,lab}.html · prototype/static/pageswitch.js · docs/mockups/live-vision.html · all edited files backed up as *.bak-2026-06-0{5,6,7}